

Exercise 1

For each of the following steps define the reply of the ECU considering that for each step the diagnostic service 19 02 08 is requested by a diagnostic tool. Consider also the following information:

- Enable condition to monitor fault = push CC button;
- Fault condition = SCB (short circuit to battery) on CC button;
- Warning lamp MIL ON 1 at the first driving cycle;
- Warning lamp heal after 2 driving cycles;
- Setting time and healing time are event based;
- DTC code "00 25" + symptom 00
- Consider bit 4 and 5 as not used
- Driving cycle: engine run
- Monitoring cycle: from key on to key off, including a driving cycle
- Consider all actions, for every step, as completely done

ID	Manoeuvre	ECU response to 19 02 08
1	Vehicle in speed limitation with cruise control	
2	Key off/key on	
3	SCB on CC button	
4	Push CC button	
5	Key off/ key on	
6	Engine run	
7	Key off/ key on	
8	Push CC button	
9	Remove SCB on CC button	

10	Push CC button	
11	Key off/ key on	
12	Push CC button	
13	Engine run, key off, key on	
14	Push CC button	
15	Key off/ key on	
16	Engine run	
17	Key off/ key on	
18	Engine run	
19	key off/ key on	
20	Engine run	
21	Push CC button	
22	SCB on CC button	
23	Push CC button	
24	Remove SCB on CC button	
25	Push CC button	
26	SCB on CC button	

27	Push CC button	
28	Engine run, key off, key on	
29	Remove SCB on cc button	
30	Push cc button	

For this exercise, use an excel file to write every ECU response, bit by bit in the case of the DTCSatusMask.